Face-Rotation BFS Nets of the Regular and Uniform Convex 4-Polytopes

Takashi Yoshino*

Abstract

We study the face-rotation BFS unfolding of regular convex 4-polytopes: a breadth-first spanning tree of the cell-adjacency graph is chosen, and each child cell is rotated about the shared polygonal face into the parent's 3-dimensional hyperplane. We prove by exhaustive computational verification that for every one of the six regular convex 4-polytopes—the 5-cell, 8-cell, 16-cell, 24-cell, 120-cell, and 600-cell—this procedure yields a valid (non-self-intersecting) 3D net from *every* choice of root cell. In total, $5+8+16+24+120+600 = 773$ root cells were verified. For the 5-, 8-, 16- and 24-cell we verify more: *every* BFS spanning tree of every root is valid, 1,118,261 trees in all, so for those four the conclusion does not depend on how ties in the traversal are broken. This extends the work of Devadoss and Harvey [3], who established validity for the 5-, 8-, and 16-cell for *all* spanning trees and left the remaining three polytopes open; we resolve the open cases for the BFS subfamily. By contrast, our result stands in sharp contrast to the apple-peel unfolding algorithm [7], for which the 120-cell produces no valid nets despite admitting 1,440 valid orderings, and the 600-cell admits no valid ordering at all. We also show that regularity is essential: running the same procedure from all 24,487 roots of the 64 convex uniform 4-polytopes leaves 61 valid from every root, one valid from none, and two valid from some roots but not others.

1 Introduction

The question of whether a convex polytope can be *unfolded* into a non-overlapping flat arrangement of its facets—a *net*—is a classical problem in discrete geometry, dating back to Dürer's depiction of polyhedral nets in the sixteenth century. In three dimensions the analogous question (does every convex polyhedron admit a non-overlapping edge unfolding?) remains open as the Dürer conjecture [4]. For four-dimensional polytopes, the facets are 3-dimensional cells and a net is an arrangement of those cells in $\mathbb{R}^3$. Early work on unfolding the 4-cube into 3D arrangements of its cubical cells was carried out by Turney [5], who enumerated all 261 topologically distinct nets of the 8-cell (hypercube).

Devadoss and Harvey [3] initiated the systematic study of ridge unfoldings of the six regular convex 4-polytopes, proving that every spanning tree of the cell-adjacency graph yields a valid net for the 4-simplex (5-cell), 4-cube (8-cell), and 4-orthoplex (16-cell). They demonstrated failure for the analogous result in dimension $n \geq 5$ for orthoplexes, but left the 24-cell, 120-cell, and 600-cell open.

In this paper we study a canonical subfamily of spanning trees, namely those produced by *breadth-first search* (BFS) from a chosen root cell, paired with a specific unfolding procedure that we call *face-rotation*. Our main result (Theorem 2) is that for every regular convex 4-polytope and every choice of root cell, this procedure yields a valid 3D net. The proof is by exhaustive computational verification. The result for the 5-, 8-, and 16-cell is a special case of [3]; the new content is the 24-cell, 120-cell, and 600-cell.

A companion paper [7] introduces the *apple-peel* algorithm, a greedy spiral/zonal ordering of cells that defines a different family of spanning trees. That algorithm completes successfully for

*`tyoshino@toyo.jp`. ORCID: 0000-0003-1756-0162.

all 1,440 starting pairs for the 120-cell, yet every resulting 3D net self-intersects. For the 600-cell the algorithm cannot even complete the ordering. The face-rotation BFS algorithm succeeds where apple-peel fails, illustrating that the existence of a valid ordering and the validity of a net are independent properties.

This paper is organized as follows. Section 2 recalls the necessary background and describes the face-rotation BFS algorithm. Section 3 states and verifies Theorem 2 and discusses the contrast with apple-peel unfolding. Section 4 runs the same procedure over the 64 convex uniform 4-polytopes and shows that the theorem does not survive the loss of regularity. Section 5 collects open problems.

2 Preliminaries

2.1 Background

There are exactly six regular convex 4-polytopes (see Coxeter [1]). Their Schläfli symbols and combinatorial data are listed in Table 1.

Table 1: The six regular convex 4-polytopes.

Polytope	Symbol	Vertices	Faces	Cells	Cell type	Cell degree
5-cell	$\{3,3,3\}$	5	10	5	tetrahedron	4
8-cell	$\{4,3,3\}$	16	24	8	cube	6
16-cell	$\{3,3,4\}$	8	32	16	tetrahedron	4
24-cell	$\{3,4,3\}$	24	96	24	octahedron	8
120-cell	$\{5,3,3\}$	600	720	120	dodecahedron	12
600-cell	$\{3,3,5\}$	120	1200	600	tetrahedron	4

Here *cell degree* denotes the number of cells adjacent to each cell (sharing a ridge, i.e., a 2-face).

Following Devadoss and Harvey [3], a *ridge unfolding* of a 4-polytope P is specified by a spanning tree T of the cell-adjacency graph $G(P)$ (nodes = cells, edges = shared ridges, i.e., shared 2-faces). The unfolding is performed inductively: a root cell is placed in some 3D hyperplane H; for each edge (p, c) in T directed away from the root, the child cell c is rigidly rotated about the shared polygon with its parent p until it lies in the same hyperplane as p. The resulting arrangement is a *net* if no two cells overlap in their interiors.

Ridge unfoldings are also accessible through interactive software. Stella4D [6] has displayed 3D nets of polychora since 2007 and lets the user reglue cells one at a time, so that in principle any spanning tree can be realized by hand. Its nets are generated to maximize symmetry and visual appeal rather than to realize a prescribed family of spanning trees, and, as its documentation states, intersections between cells are ignored. Whether an unfolding is free of self-intersection—the question studied here—is therefore outside the scope of such tools.

Definition 1 (BFS spanning tree). *Let $G(P)$ be the cell-adjacency graph of a 4-polytope P, let r be a root cell, and write L_k for the set of cells at distance k from r in $G(P)$. A* BFS spanning tree *rooted at r is a spanning tree in which the parent of each cell $v \neq r$ is one of its neighbors in $L_{d(v)-1}$, where $d(v)$ is the distance from r to v. The layers L_k are determined by P and r alone; a BFS tree is exactly a choice of such a parent for each cell, and different choices give different trees. Equivalently, a spanning tree rooted at r is a BFS tree if and only if every path in it from r is a shortest path in $G(P)$. We use the name for this family of trees, not for any one traversal: the definition above mentions no order in which cells are visited. Note also that $G(P)$ itself does not depend on the root: the root selects a tree in it.*

Lemma 1 (What a traversal can reach)**.** *A BFS tree rooted at r is produced by some breadth-first traversal if and only if there is a total order $\prec$ on the cells with*

$$p(v) \;=\; \min_{\prec}\bigl(N(v) \cap L_{d(v)-1}\bigr) \qquad \textit{for every } v \neq r,$$

and such an order exists if and only if the digraph with an edge $p(v) \to q$ for every other candidate q of every v is acyclic.

Proof. A traversal handles the cells of a layer in the order it dequeues them, and v is claimed by the first of its candidates to be dequeued; that dequeue order is a total order, which gives one direction. Conversely, given $\prec$, process the layers in turn and each layer in increasing $\prec$: then v is claimed by its $\prec$-least candidate. The order must satisfy $p(v) \prec q$ for every candidate $q \neq p(v)$ of every v, and a total order meeting a set of such constraints exists exactly when they have no cycle. Since all candidates of v lie in $L_{d(v)-1}$, the digraph splits by layer. □

Remark 1 (The family is larger than what a traversal reaches)**.** *Lemma 1 makes the gap between the two readings of "BFS tree" measurable, and it is wide. Table 2 counts, for the polytopes of part 1 of Theorem 2, how many of the trees allowed by Definition 1 a traversal can reach. The counts are the same from every root, as cell-transitivity requires.*

Table 2: BFS trees per root, and how many of them a breadth-first traversal can produce.

Polytope	BFS trees	Reachable	
5-cell	1	1	100%
8-cell	6	6	100%
16-cell	20,736	4,896	23.6%
24-cell	32,768	2,592	7.9%

The obstruction is not the obvious one. If two cells of a layer share two candidates, no traversal can give them different parents from that pair; but the 16-cell has no such pair from any root and still leaves three quarters of its trees unreachable, because the constraints can close a cycle without any two cells sharing two candidates—v_1 taking a over b, v_2 taking b over c and v_3 taking c over a is already impossible.

We take Definition 1 as the definition, so part 1 of Theorem 2 is correspondingly stronger. For the 24-cell it verifies 786,432 *trees of which only* 62,208 *are reachable by any traversal: the other* 724,224 *are trees no implementation of breadth-first search will ever hand you.*

Remark 2 (The tree we compute)**.** *Selecting the parents requires a rule, and unless stated otherwise the tree we compute, written $\mathcal{T}_r$, is the one produced by the textbook BFS loop: cells are held in a FIFO queue, the queue is initialized with r, and each time a cell is dequeued its neighbors are scanned in increasing cell index, every neighbor not yet seen being marked, given the dequeued cell as its parent, and appended to the queue. The parent of v is thus whichever of its candidates entered the queue first.*

We spell this out because the rule is less innocent than it looks. The only arbitrary ingredient is the index order in which neighbors are scanned, but the queue order that decides parentage is derived from it recursively: cells at distance $d-1$ are dequeued in the order their own parents were, so the induced order is lexicographic along the discovery path rather than by index. It is therefore not the same as taking the smallest-indexed candidate. On the two polytopes of Section 4 for which the choice turns out to matter, the two rules disagree on 55% *and* 25% *of the cells that have more than one candidate. Note also that a traversal is not even able to reach every tree Definition 1 allows, by Remark 1; the trees enumerated for part 1 of Theorem 2 are chosen directly from the definition, not run through a traversal. Neither rule has any geometric content. For the polytopes of part 1 of Theorem 2 none of this matters, since* every *BFS tree is valid there; where it matters is Section 4.*

2.2 Face-Rotation BFS Algorithm

Given a 4-polytope P with vertex set $V \subset \mathbb{R}^4$, cell set C, and a root cell $r \in C$, the algorithm proceeds as follows.

Step 1: Face-up rotation. Let $\bar{V} = \{v - \bar{v} : v \in V\}$ where $\bar{v}$ is the centroid of V. Let $c_r = \mathrm{centroid}(\{v \in \bar{V} : v \text{ is a vertex of } r\})$. Apply the rotation $R \in \mathrm{SO}(4)$ that maps $c_r/\|c_r\|$ to the unit vector $\hat{w} = (0,0,0,1)$. (If $c_r/\|c_r\| = \hat{w}$ we take $R = I$; if $c_r/\|c_r\| = -\hat{w}$ we negate all coordinates.) This *face-up* rotation ensures that the root cell's centroid lies on the positive w-axis, making subsequent unfolding into the hyperplane $\{w = \mathrm{const}\}$ numerically stable.

Step 2: BFS traversal. Maintain a FIFO queue Q initialized with the root cell r, and a visited set $\mathrm{Vis} = \{r\}$ with $T_r = \mathrm{id}$. While Q is non-empty, dequeue the front cell p; for each neighbor c of p in $G(P)$, taken in the order they appear in the adjacency list, if $c \notin \mathrm{Vis}$: add c to Vis, enqueue c, and compute the cumulative 4D affine transform T_c by the procedure in Algorithm 1. The algorithm terminates when Q is empty (no candidates remain); since $G(P)$ is connected for every regular 4-polytope, all cells are visited. The order in which neighbors are added to Q constitutes the tie-breaking rule of Definition 1 and determines which BFS spanning tree is produced. In our implementation, ties are broken by the order in which neighbors appear in the adjacency list of the input data.

Step 3: Projection. Apply each T_c to the vertices of cell c, obtaining $\{\, T_c(v) \in \mathbb{R}^4 : v \in c \,\}$. After Step 2, the w-coordinate of every transformed vertex satisfies $|w - w_0| \le \varepsilon_w$ for a common value w_0 and numerical tolerance $\varepsilon_w \approx 10^{-13}$. Discard the w-coordinate; the resulting (x, y, z)-coordinates form the 3D net.

Algorithm 1 $\textsc{UnfoldTransform}(p, c, T_p)$

Require: parent cell p with cumulative transform T_p; child cell c
Ensure: cumulative transform T_c for c

Let $f_1, \dots, f_k$ be the vertices of the shared face of p and c (in the original 4D frame)
$\tilde{f}_i \leftarrow T_p(f_i)$ for all i ▷ shared face in parent's unfolded frame
$c_f \leftarrow \frac{1}{k} \sum_i \tilde{f}_i$ ▷ face centroid
Compute orthonormal basis $\{u_1, u_2\}$ of the face plane via Gram-Schmidt on $\{\tilde{f}_i - c_f\}$
$n_p \leftarrow$ component of $(T_p(\mathrm{centroid}(p)) - c_f)$ orthogonal to $\{u_1, u_2\}$, normalized
$n_c \leftarrow$ component of $(T_p(\mathrm{centroid}(c)) - c_f)$ orthogonal to $\{u_1, u_2\}$, normalized
$\theta \leftarrow \arccos\big(\max(-1,\ \min(1,\ n_c \cdot (-n_p)))\big)$
$e_1 \leftarrow n_c$; $\quad e_2 \leftarrow \mathrm{normalize}(-n_p - (n_c \cdot (-n_p))\, n_c)$ ▷ if degenerate, use any vector $\perp \{u_1, u_2, n_c\}$
$R \leftarrow I_4 + (\cos\theta - 1)(e_1 e_1^\top + e_2 e_2^\top) + \sin\theta (e_2 e_1^\top - e_1 e_2^\top)$ ▷ 4D rotation in the plane $\{e_1, e_2\}$
return $\big(R,\ c_f - R\, c_f\big) \circ T_p$ ▷ compose affine maps

A naive implementation using *closed-region* membership (e.g. Mathematica's `RegionMember`) incorrectly classifies cells as overlapping when they share only a face, edge, or vertex in the net. This is particularly acute for pointed cells such as tetrahedra and octahedra, which frequently share edges and vertices with cells not connected by an edge of $\mathcal{T}_r$.

We use the following *volume-based* test, which detects only interior overlaps.

Remark 3. *Two cells not connected by an edge of $\mathcal{T}_r$ may share a face, edge, or vertex in the unfolded net without overlapping in their interiors. A correct validity test must detect only* interior *overlaps.*

Definition 2 (Valid net)**.** *A net is* valid *if for every pair of cells* (i, j) *not connected by an edge of* $\mathcal{T}_r$*,*

$$\mathrm{Vol}_3\big(\mathrm{conv}(c_i) \cap \mathrm{conv}(c_j)\big) \;\leq\; \varepsilon_{\mathrm{vol}}$$

where $\mathrm{conv}(c_i)$ *denotes the convex hull of cell* i*'s vertices in 3D, and* $\varepsilon_{\mathrm{vol}} = 10^{-8}$.

In practice we first apply a bounding-box filter to exclude pairs that clearly cannot overlap, then compute `RegionMeasure[RegionIntersection[`h_i`, `h_j`]]` in Mathematica for the remaining candidate pairs, where h_i denotes the convex hull mesh of cell i's vertices in 3D.

Remark 4. *Using closed-region membership instead of volume intersection would report the 16-cell, 24-cell, and 600-cell as invalid (false positives due to boundary contact), contradicting the correct result. The volume threshold* $\varepsilon_{\mathrm{vol}} = 10^{-8}$ *is well below the smallest possible cell volume (even for the 600-cell, where each tetrahedral cell has volume* $\approx 3 \times 10^{-3}$ *at the scale used), so no true interior overlap is missed.*

3 Results

3.1 Main result

By Definition 1 the BFS trees rooted at r are in bijection with the parent choices, so there are exactly

$$\prod_{v \neq r} \big|\, N(v) \cap L_{d(v)-1} \,\big|$$

of them. This count mentions no rule, and lets us enumerate or sample the trees without reference to one. For the 5-cell the product is 1: there is only one BFS tree per root and a tie-breaking rule has nothing to decide. For the 8-, 16- and 24-cell it is 6, 20,736 and 32,768; for the 120- and 600-cell it is about 1.8×10^{44} and 1.8×10^{65}.

Theorem 2 (Main)**.** *Let* P *be a regular convex* 4*-polytope and* r *a cell of* P*.*

1. *If* P *is the 5-cell, 8-cell, 16-cell or 24-cell, then* every *BFS spanning tree rooted at* r *yields a valid net. In particular the conclusion does not depend on the tie-breaking rule.*
2. *If* P *is the 120-cell or the 600-cell, then the tree* $\mathcal{T}_r$ *of Remark 2 yields a valid net.*

Proof. Exhaustive computational verification. For each of the six polytopes we loaded the combinatorial data (vertices, faces, cells) from the data files available in the repository, unfolded from every root cell, and applied the volume-based intersection test (Definition 2) to every pair of cells not joined by an edge of the tree in question.

For part 1 we enumerated all BFS trees of every root: 5, 48, 331,776 and 786,432 trees for the 5-, 8-, 16- and 24-cell respectively. All are valid. For the 5-, 8- and 16-cell this also follows from [3], which covers all spanning trees; the 24-cell is new.

For part 2 the number of BFS trees puts enumeration out of reach, and we verified the tree $\mathcal{T}_r$ for each of the 120 and 600 roots. As evidence that the outcome is not an artifact of the rule, we additionally drew 20 BFS trees uniformly at random per root— 2,400 and 12,000 trees—and found all of them valid as well. Across all six polytopes 1,132,661 trees were checked without a single self-intersection.

Table 3 summarizes the results, and Figure 1 shows representative nets for each polytope. The w-coordinate spread after unfolding was at most 4×10^{-13} in all cases, confirming that all cells lie in a common 3D hyperplane to numerical precision. □

Table 3: Comparison of face-rotation BFS and apple-peel RZ unfolding. "Valid orderings" = fraction of (C_1, C_2) pairs for which apple-peel visits all cells; "Valid nets" = fraction of valid orderings that produce a non-self-intersecting 3D net.

		Face-rotation BFS	Apple-peel RZ [7]	
Polytope	Cells	Valid (all roots)	Valid orderings	Valid nets
5-cell	5	5/5	20/20 (100%)	20/20 (100%)
8-cell	8	8/8	48/48 (100%)	48/48 (100%)
16-cell	16	16/16	20/64 (31%)	20/20 (100%)
24-cell	24	24/24	192/192 (100%)	192/192 (100%)
120-cell	120	120/120	1440/1440 (100%)	**0/1440 (0%)**
600-cell	600	600/600	**0/2400 (0%)**	—

Remark 5. *For the 5-, 8-, and 16-cell, Theorem 2 is a special case of the stronger result of Devadoss and Harvey [3], who proved that* all *spanning trees of the cell-adjacency graph (not just BFS trees) yield valid nets. The new content of Theorem 2 is therefore the 24-cell, 120-cell, and 600-cell.*

Remark 6 (On the number of verifications)**.** *Each regular convex* 4*-polytope has a symmetry group acting transitively on cells. If the BFS algorithm uses a geometrically equivariant tie-breaking rule, then for any symmetry g mapping root r to root r', the net from r' is the image of the net from r under g, and validity is preserved by the rigidity of g. In that case, verification of a* single *root cell per polytope would suffice to establish Theorem 2. Our implementation uses index-based tie-breaking in the BFS queue, which is not equivariant under the symmetry group; we therefore verified all* 773 *roots independently. In either case, Theorem 2 guarantees the* existence *of a valid face-rotation BFS net from every root cell under this tie-breaking rule. Whether validity holds for every possible tie-breaking rule remains open.*

3.2 Contrast with Apple-Peel Unfolding

The apple-peel algorithm [7] constructs a spanning tree by greedily selecting the next cell according to a global spiral (RS) or zonal (RZ) criterion, without the layer-by-layer structure of BFS. Table 3 compares the two algorithms.

The most striking contrast occurs for the 120-cell and 600-cell:

- **120-cell**: Apple-peel produces 1,440 valid orderings (all (C_1, C_2) pairs succeed), yet every resulting 3D net self-intersects [7]. Face-rotation BFS yields a valid net from every root.
- **600-cell**: Apple-peel cannot complete any ordering (structural impossibility due to the 4-regular cell-adjacency graph). Face-rotation BFS yields a valid net from every root.

These cases demonstrate that *the existence of a valid ordering and the validity of the resulting 3D net are independent properties.* A spanning tree that is well-suited for the ordering problem need not yield a geometrically valid net, and vice versa.

4 The Uniform Case

Theorem 2 does not extend to the convex uniform 4-polytopes. Excluding the two infinite prismatic families, there are exactly 64 of these; unlike the enumeration of all uniform 4-polytopes, which is still open, this list is known to be complete, the exceptional non-Wythoffian member being the grand antiprism of Conway and Guy [2]. We generated all 64 and ran the face-rotation BFS unfolding from every cell, 24,487 roots in all.

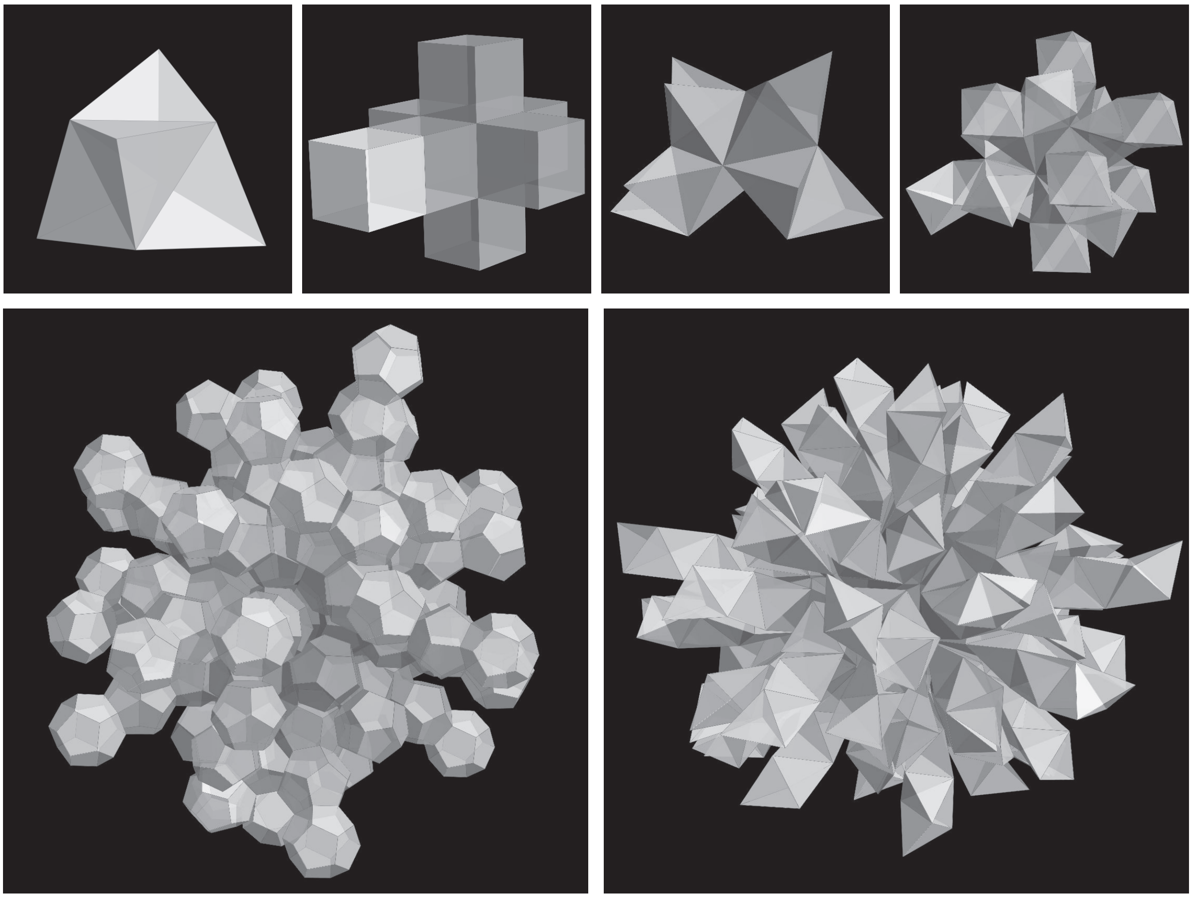


Figure 1: Face-rotation BFS nets of the six regular convex 4-polytopes. Top row (left to right): 5-cell, 8-cell, 16-cell, 24-cell. Bottom row: 120-cell, 600-cell. Shading indicates BFS layer depth (dark = root cell, light = periphery). STL files of all six nets are available in the repository (see Code and Data Availability).

Construction. The 45 distinct Wythoffian members of the A_4, B_4, F_4 and H_4 families were built from their Coxeter diagrams: the group is generated as the closure of the four mirror reflections, the vertices are the orbit of the point whose distance to each ringed mirror is 1 and to each unringed mirror 0 (the edge generated by mirror i has length twice that distance, so equal distances force equal edge lengths), and the d-faces are the orbits of the rank-d parabolic subgroups. The snub 24-cell and the grand antiprism are not Wythoffian in this sense, but both arise as diminishings of the 600-cell: delete an inscribed 24-cell, respectively two orthogonal decagonal rings, and take the convex hull. In each case the surviving cells are the 600-cell tetrahedra avoiding every deleted vertex, together with one cell per deleted vertex, namely the hull of its remaining neighbors. The 17 prisms are the prisms over the 5 Platonic and 13 Archimedean solids, with lateral edge length equal to the edge length of the base, minus the cube prism, which is the tesseract and already regular. Each construction was checked against the group-theoretic predictions for the vertex and cell counts, the Euler relation $V-E+F-C=0$, the requirement that every ridge lie in exactly two cells, and vertex-transitivity of the vertex degree.

Results. The outcome splits three ways, as summarized in Table 4.

Table 4: Face-rotation BFS nets of the 64 convex uniform 4-polytopes, over all root cells.

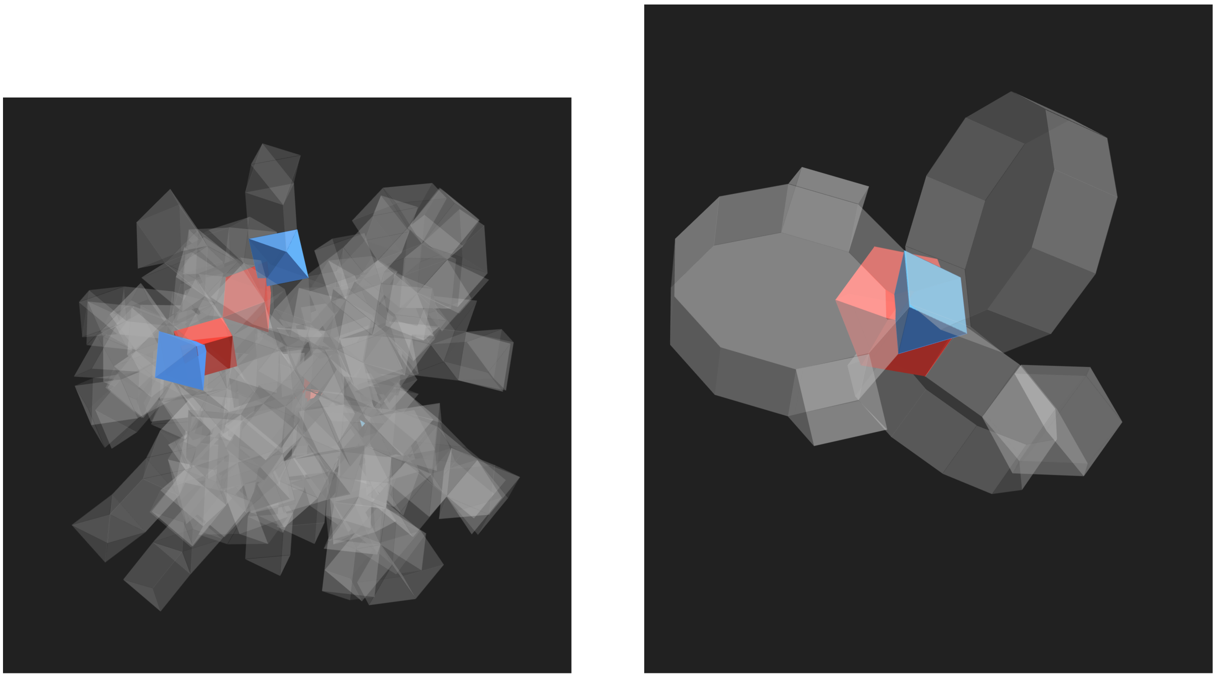
Family	Polytopes	Roots	All valid	Root-dependent	None valid
A_4, B_4, F_4	30	1,893	29	0	1
H_4	15	21,600	13	2	0
Snub 24-cell, grand antiprism	2	464	2	0	0
Prisms	17	530	17	0	0
Total	64	24,487	61	2	1

Sixty-one of the 64 yield a valid net from every root. The runcinated 24-cell $t_{0,3}\{3,4,3\}$ yields one from no root: all 240 of its unfoldings self-intersect, in between 3 and 23 pairs of cells. The remaining two are *root-dependent*: the runcitruncated 600-cell $t_{0,2,3}\{5,3,3\}$ succeeds from 47 of its 2,640 roots and the runcitruncated 120-cell $t_{0,1,3}\{5,3,3\}$ from 2,570 of 2,640. Figure 2 shows one failure of each kind.

The two root-dependent cases are the informative ones. For every regular 4-polytope the choice of root is immaterial, and it is tempting to read that as a consequence of cell-transitivity. The uniform case shows that once cell-transitivity is dropped, validity genuinely depends on where the unfolding starts: admitting *some* valid face-rotation BFS net and admitting one from *every* root are different properties, and Theorem 2 establishes the stronger one only in the regular case.

Cell degree does not separate the three outcomes, in either direction. The grand antiprism and the snub 24-cell both contain cells of degree 4—the same poverty of connection that makes the 600-cell impossible for apple-peel unfolding—yet both are valid from every root, while the runcinated 24-cell has minimum cell degree 5 and is valid from none. Nor does high degree help: every prism has two cells adjacent to all of the others, up to degree 92 for the prism over the snub dodecahedron, and all 17 are valid from every root, whereas the root-dependent runcitruncated 120-cell also carries cells of degree 32.

Caveats. The BFS tree depends on the tie-breaking rule of Definition 1. For a cell-transitive polytope the rule is immaterial, but the uniform polytopes have several cell orbits, so the counts 47 and 2,570 above are those of one fixed rule (cell index order); another rule may change them. For the polytopes with more than a few hundred cells the intersection test was carried out with a separating-axis implementation rather than by measuring the volume of the intersection; the



(a) (b)

Figure 2: The two ways the unfolding fails. The panels are at deliberately different scales, because the claims they carry are about different things: (a) is about a polytope, (b) about a single root. (a) The runcinated 24-cell at its *best* root, drawn whole: 3 of its 240 cells intersect a partner, the fewest over all 240 roots, and no root does better than 3. (b) The runcitruncated 120-cell at one of its 70 failing roots, drawn as a close-up of the single intersecting pair together with its neighbors; the other 2,570 roots are fine. In both panels the two members of an intersecting pair are drawn in different colors.

two were checked to return identical verdicts, and identical numbers of intersecting pairs, on every one of the 1,893 roots of the 30 polytopes of the A_4, B_4 and F_4 families, and again on a sample of roots of the two root-dependent cases, where the verdict is also stable as the separation tolerance ranges over 10^{-4} to 10^{-10}.

5 Open Problems

The main open question raised by this work is:

> *Why does face-rotation BFS always produce a valid net for regular 4-polytopes? Is there a theoretical proof that does not rely on computer enumeration?*

A second open problem is whether Theorem 2 extends to all spanning trees (not just BFS) for the 24-cell, 120-cell, and 600-cell, as was proved for the 5-, 8-, and 16-cell by [3]. Within the BFS family the 24-cell is settled by part 1 of the theorem, so what remains there is the step from BFS trees to arbitrary spanning trees. For the 120- and 600-cell even the BFS family is open: with 1.8×10^{44} and 1.8×10^{65} trees per root, enumeration is hopeless and a proof would have to be structural rather than computational.

A third question is the characterization left open by Section 4. What distinguishes the 61 uniform 4-polytopes that unfold from every root, from the two that unfold from some roots only, and from the runcinated 24-cell, which unfolds from none? As noted there, cell degree does not decide it, and we have no candidate invariant that does. A related question is how far the counts for the two root-dependent cases depend on the tie-breaking rule: whether some rule makes them valid from every root, or whether root-dependence survives every choice.

Finally, extensions to higher dimensions are open as well.

Code and Data Availability

All Mathematica scripts implementing the face-rotation BFS algorithm, the volume-based intersection test, and the visualization are available at `https://github.com/takashi-randomwalker/apple-peel-4d`. The same repository contains the constructions used for Section 4—the Wythoff generator, the two diminishings of the 600-cell, and the prisms—and the separating-axis implementation of the intersection test used for the larger polytopes, together with its agreement check against the volume-based test. It also contains the combinatorial data of all 64 convex uniform 4-polytopes, the nets exported as STL and OFF, and the verdict for each of the 24,487 individual roots. STL files of the face-rotation BFS nets for all six polytopes (root cell 1) are also provided in the same repository.

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